USER		
INT	Id	PK
VARCHAR	Username	
VARCHAR	Email	
VARCHAR	password_hash	
DATETIME	created_at	

POSTS		
INT	Id	PK
INT	user_id	FK
VARCHAR	title	
TEXT	description	
VARCHAR	Image_url	

COMMENTS		
INT	id	PK
INT	post_id	FK
INT	user_id	FK
TEXT	content	
DATETIME	Created_at	

LIKES		
INT	id	PK
INT	post_id	FK
INT	user_id	FK
DATETIME	Created_at	

Exercise 1 – Blogging Application Database

Logical Model

Entities and Attributes

Users: user_id (PK), username, email, password, created_at

Posts: post_id (PK), user_id (FK), title, description, image_url, created_at

Comments: comment_id (PK), post_id (FK), user_id (FK), comment_text, created_at

Likes: like_id (PK), post_id (FK), user_id (FK)

Relationships

One user can create many posts (one-to-many).
One post can have many comments (one-to-many).
One user can write many comments (one-to-many).
Users and posts have a many-to-many relationship through likes, because one user can like many posts and one post can be liked by many users. The Likes table is the linking table between them.

Normalization

I did not store a like count in the Posts table, because it can be calculated by counting the rows in Likes.
I did not store the author's name in Posts. I used user_id instead, so the user's information is only stored once.
The image is stored as a URL instead of the actual image file.

Physical Model (MySQL)

Users
user_id INT, Primary Key, Auto Increment
username VARCHAR(50), Not Null, Unique
email VARCHAR(100), Not Null, Unique
password VARCHAR(255), Not Null
created_at DATETIME

Posts
post_id INT, Primary Key, Auto Increment
user_id INT, Foreign Key to Users, Not Null
title VARCHAR(100), Not Null
description TEXT

image_url VARCHAR(255)
created_at DATETIME

Comments
comment_id INT, Primary Key, Auto Increment
post_id INT, Foreign Key to Posts, Not Null
user_id INT, Foreign Key to Users, Not Null
comment_text TEXT, Not Null
created_at DATETIME

Likes
like_id INT, Primary Key, Auto Increment
post_id INT, Foreign Key to Posts, Not Null
user_id INT, Foreign Key to Users, Not Null
The combination of post_id and user_id is unique, so a user can only like a post once.